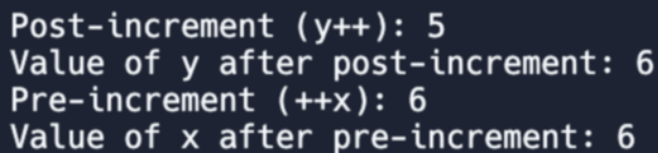


//Pre vs Post Increment in C++

```
#include <iostream>
using namespace std;

int main(){
    int x = 5;
    int y = 5;
    // Post-increment: value used, then incremented
    cout << "Post-increment (y++): " << y++ << endl; // Prints 5, y becomes 6
    cout << "Value of y after post-increment: " << y << endl; // y is now 6
    // Pre-increment: value incremented, then used
    cout << "Pre-increment (++x): " << ++x << endl; // Prints 6, x becomes 6
    cout << "Value of x after pre-increment: " << x << endl; // x is now 6
}
```

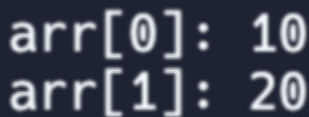


```
Post-increment (y++): 5
Value of y after post-increment: 6
Pre-increment (++x): 6
Value of x after pre-increment: 6
```

//Pointer and array relationships

```
#include <iostream>
using namespace std;

int main(){
    int arr[3] = {10, 20, 30};
    int *ptr = arr; // Points to arr[0]
    cout << "arr[0]: " << *ptr << endl; // 10
    cout << "arr[1]: " << *(ptr + 1) << endl; // 20
}
```



```
arr[0]: 10
arr[1]: 20
```

//Pointer arithmetic

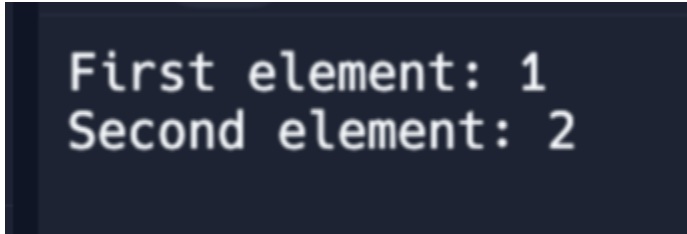
```
#include <iostream>
using namespace std;

int main(){
    int arr[] = {1, 2, 3, 4, 5};
```

```

int *ptr = arr;    cout << "First element: " << *ptr << endl; // 1
ptr++;
cout << "Second element: " << *ptr << endl; // 2
}

```



```

First element: 1
Second element: 2

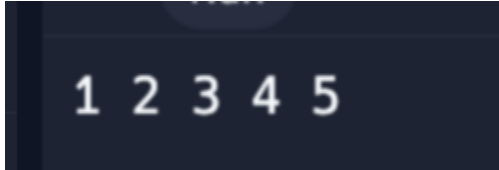
```

//How to pass arrays as pointers to functions

```

#include <iostream>
using namespace std;
void printArray(int* ptr, int size) {
    for (int i = 0; i < size; i++) {
        cout << *(ptr + i) << " ";
    }
}
int main() {
    int arr[] = {1, 2, 3, 4, 5};
    printArray(arr, 5);
    return 0;
}

```



```

1 2 3 4 5

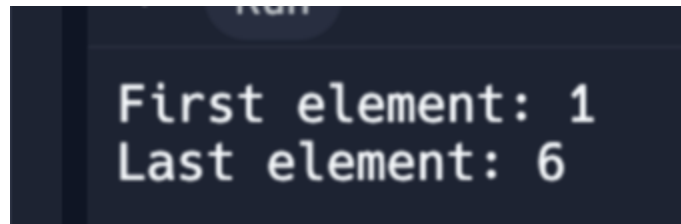
```

//2D Arrays and pointers

```

#include <iostream>
using namespace std;
int main() {
    int arr[2][3] = {{1, 2, 3}, {4, 5, 6}};
    int (*p)[3] = arr;
    cout << "First element: " << (*(p + 0) + 0) << endl;
    cout << "Last element: " << (*(p + 1) + 2) << endl;
    return 0;
}

```



First element: 1
Last element: 6

//Dynamic memory allocation with pointers

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

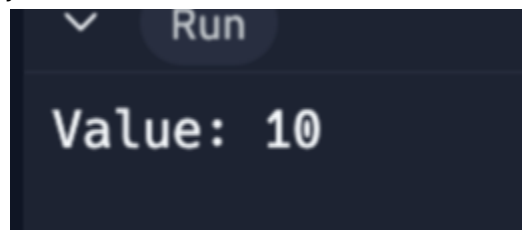
```
    int *ptr = new int(10);
```

```
    cout << "Value: " << *ptr << endl;
```

```
    delete ptr;
```

```
    return 0;
```

```
}
```



Value: 10